Linxiao Wu

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EDUCATION

Columbia University — New York, NY
M.S. in Electrical Engineering | GPA: 4.0/4.0 — Expected Dec 2026
Sun Yat-sen University — Guangzhou, China
B.Eng. in Microelectronics Science and Engineering | GPA: 3.7/4.0 — Jun 2025
Honors: Sun Yat-sen University Scholarship; Outstanding Undergraduate Thesis Award

EXPERIENCE

WingSemi Technology — Shanghai Zhangjiang, China
CPU IP Design Intern — Jun 2026 - Sep 2026

- Contributed to an RVA23-compliant RISC-V CPU IP, focusing on VFPU design and integration within the VPU.
- Owned the VFCVT module, implementing conversions among FP64, FP32, FP16, BF16, and integer formats for RVV instructions.
- Designed a three-stage pipeline with shared multi-precision datapaths, lane-level parallelism, configurable DLEN, and PPA-driven hardware reuse.
- Supported VFPU/VCTRL integration, vector micro-op control, mask/lane management, valid/stall/flush handling, and pipeline forwarding.
- Performed RTL synthesis and timing analysis in Synopsys Design Compiler and optimized critical datapaths.

Guangzhou CanSemi Technology Inc. — Guangzhou, China
TCAD / Device Engineering Intern — Jul 2024 - Sep 2024

- Built and calibrated a reusable 32 V PMOS TCAD model against foundry process data; analyzed wafer electrical and WAT data.
- Simulated IGBT device structures and supported patent-related device development.

PROJECTS

Apple-Sponsored Gesture-Recognition CNN Accelerator ASIC — Feb 2026 - Present

- Designed a fully hardware INT8 CNN for 64 x 64 grayscale gesture recognition; taped out in TSMC 65 nm in July 2026.
- Built a 64-PE INT8 systolic array with approximately 464K MACs per inference and 8 KB on-chip SRAM.
- Completed RTL, DFT scan-chain design, Genus synthesis, Innovus place and route, and post-layout verification; evaluated quantization in Python and MATLAB.

P6-Based Out-of-Order RISC-V Processor — Mar 2026 - Jun 2026

- Designed two-wide fetch/dispatch/retirement with single-issue out-of-order execution and in-order commit.
- Implemented a 16-entry ROB, 8-entry reservation station, map table, and CDB for ROB-tag-based register renaming and issue.
- Developed a 5-bit-GHR, 32-entry BHT/BTB gshare predictor with speculative updates and ROB-snapshot recovery.
- Implemented split LQ/SQ and a two-way non-blocking D-cache with store-to-load forwarding, four-entry MSHR capacity, and a four-entry victim cache.

N-Body Accelerator Formal Verification — Oct 2025 - Dec 2025

- Authored SystemVerilog assertions, assumptions, and cover properties in Cadence JasperGold for FSM, shift-register, and pointer checks.
- Found an illegal FSM transition missed by simulation and a one-cycle pointer-range violation.

64-Tap 16-Bit FIR Digital Filter — Oct 2025 - Dec 2025

- Designed a fixed-point FIR core with a multiply-accumulate datapath, CMEM/IMEM, and control FSM.
- Used an asynchronous FIFO for CDC between the 10 kHz input domain and the high-speed FIR core.

TECHNICAL SKILLS

EDA Tools: Synopsys Design Compiler, PrimeTime; Cadence Genus, Innovus, Xcelium; JasperGold, ModelSim, Verdi
Programming: Verilog, SystemVerilog, Python (NumPy, PyTorch), C, MATLAB
Focus Areas: ASIC design, RISC-V CPU/VPU architecture, CNN hardware accelerators